Abstract

This project is a challenge to implement the QUIC transport protocol from scratch in C. The main goal is to develop a practical understanding of how QUIC works by implementing its core components and mechanisms. The project will be developed incrementally and tested throughout development.

1 Change Log

- **August 2026** – Project started. Initial project structure and documentation created.

QUICC-Ashburn

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2 Introduction

This project is a challenge project undertaken with the goal of developing a QUIC implementation from scratch in the C programming language. The purpose of the project is to gain a deeper understanding of the QUIC protocol, its underlying mechanisms, and the challenges involved in implementing a modern transport protocol.

The implementation will aim to recreate the core functionality of QUIC, including packet handling, connection establishment, streams, flow control, loss detection, and congestion control. Rather than relying on an existing QUIC implementation, the project will be developed independently to provide a practical understanding of how the protocol operates at the implementation level.

This project is primarily intended as a learning and experimentation exercise. It is not initially intended to be a production-ready or fully standards-compliant implementation. As development progresses, the implementation will be tested against the QUIC specification and, where practical, existing implementations to evaluate its correctness and interoperability.